

# Execution of Modal Triplet Theory I: Gauge, HYM, Threshold, and Axion Status after True-SM Closure

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## Abstract

We replace the obsolete few-TeV Calabi–Yau benchmark with the current audited execution status of the MTT gauge, Hermitian Yang–Mills (HYM), threshold, and axion sectors. The selected gauge profile is transported by SMDR v1.3 to  $Q = M_t$  and enters an eight-row positive-definite precision object. Embedded renormalized-Standard-Model equivalence closes at the adopted one-shared-physical-primitive/profile standard. On the selected  $q79/F/m1$  rank-two bundle, the literal Cech–HYM witness closes, including the finite cocycle and continuum tail certificates. The internal  $K_{\text{threshold}}$  response ledger also closes, but physical threshold/mass rows remain admitted profile data rather than strict MTT-emitted values. The old fitted Kahler moduli, exceptional-cycle thresholds, and axion normalizations are withdrawn. Strong CP remains open because the complete  $E_6$  color anomaly cancels and no selected flux/threshold axion-current map has yet supplied a nonzero effective anomaly.

## Revision note for this edition

**Supersedes.** *Execution of Modal Triplet Theory I: Gauge, Axion, and Threshold Sectors*, version 2.

**Reason.** The former execution relied on a fitted few-TeV Calabi–Yau benchmark, obsolete moduli and threshold values, and axion normalizations not selected by the current q79 source.

**Resolution.** Version 3 replaces that benchmark with the SMDR precision profile, the certified q79/F/m1 Cech–HYM witness, the closed internal threshold ledger, and an explicit strong-CP source audit.

**Retained result.** Gauge/HYM and threshold structures support embedded renormalized-SM equivalence at the adopted profile standard.

**Remaining boundary.** Strict emitted physical threshold values and a selected nonzero effective color-anomaly map remain stronger upgrades.

## 1 Status standard and authority

This paper reports two distinct closure standards:

1. **Adopted profile standard.** MTT embeds the renormalized Standard Model on the selected branch, with one shared physical primitive and measured common-scheme profile rows. The final audit closes 12/12 obligations.
2. **Strict source standard.** Every numerical value must be emitted from selected MTT source geometry without empirical profile rows. This stronger zero-primitive/no-knob standard remains open.

The distinction is essential: closure at the first standard must not be described as closure at the second.

The old Tier-3/Tier-4 calculation used a gauge crossing near 5 TeV to fit Kahler ratios, thresholds, and axion data. That crossing is withdrawn. None of its downstream numerical geometry is retained as current authority.

## 2 Selected gauge and precision execution

The selected common-scheme point is

$$Q = M_t = 172.5590883453979 \text{ GeV}.$$

The gauge rows are

$$g_Y = 0.3585945042 \pm 0.0000307251, \tag{2.1}$$

$$g_2 = 0.6475986708 \pm 0.0000287665, \tag{2.2}$$

$$g_3 = 1.163427409 \pm 0.004036156, \tag{2.3}$$

with  $g_1 = \sqrt{5/3} g_Y$  when GUT normalization is required. These are measured-profile coordinates transported by the selected multi-loop Standard Model pipeline; they are not first-principles MTT predictions.

The full selected precision map transports 15 common source coordinates into eight  $\overline{\text{MS}}$  output rows. Its covariance is positive definite, contains all 36 symmetric entries, and contains all 15 BCT–WZH cross-block entries. The declared reproducible baseline is the diagonal measured-input profile because no public official joint likelihood for all 15 coordinates was identified.

## 3 HYM and bundle execution

The selected literal bundle result is stronger than the legacy string-lift ansatz. On the selected  $q79/F/m1$  rank-two bundle:

- the finite Cech witness contains 81 table entries and all 729 cocycle triples;
- the projected HYM solution has residual  $8.21 \times 10^{-13}$ , coercivity margin 26.02, and error indicator  $3.15 \times 10^{-14}$ ; and
- the weighted-theta Fourier-tail/Wiener certificate has  $Z = 0.38508$  and  $Y + Zr = 0.00932703 < r = 0.01$ .

Together these close the selected rank-two literal Cech–HYM witness families 2/2, including continuum existence and local uniqueness in the certified ball. They do not prove uniqueness over all HYM branches or rank-three sector/operator transfer.

This  $q79/\text{Fu}$ –Yau bundle result is not an identification with the auxiliary  $L(3, 1) \times \text{Nil}_3$  overlap model used in the Theta papers. The shared circle is common  $U(1)$  phase/holonomy data in the  $q79$  carrier and is not an extra Cartesian factor. A same-source world-in-world/strain-to- $q79$  intertwining theorem remains a separate globalization target.

## 4 Threshold execution

Two threshold layers must be kept separate.

## 4.1 Internal response layer

The selected internal  $K_{\text{threshold}}$  ledger closes 10/10. The direct  $K_{\text{threshold}}.\Omega_H.\lambda$  row is locked, and the charged  $K$ -to- $\Omega$  construction is available through the selected source rule

$$\Omega_i^{\text{src}} = D_{\text{fin}}[\text{class}(i)] K_{\text{threshold}}.\Omega_i e^{-2\pi n_i}.$$

This is a closure statement about the finite internal response machinery.

## 4.2 Physical matching layer

Seven physical threshold rows and three mass-scheme rows are closed only at the admitted-external/profile tier. Their strict MTT-emitted value count remains zero. The selected multi-loop transport and covariance therefore make the profile execution reproducible, but they do not convert measured matching rows into source-derived constants.

The legacy one-loop bulk logarithm, exceptional-cycle vector, and minimum-norm threshold fit are withdrawn as numerical results. They may be retained only as historical ansatz templates for a future source-selected compactification.

## 5 Axion and strong-CP status

The current corpus supports a conditional Peccei–Quinn mechanism and selected axion decay-constant ratios. It does not select the absolute axion normalization or solve strong CP.

For the complete  $E_6$   $Q_\psi$  current, the exact color-anomaly audit gives

$$\mathcal{A}_{\text{matter}} = +12, \quad \mathcal{A}_{\text{complete-27 exotics}} = -12, \quad \mathcal{A}_{\text{total}} = 0.$$

The matter-only/singlet diagnostic  $N_{\text{DW}} = 3$  is not the anomaly of the complete spectrum. A strong-CP theorem therefore requires a selected flux/threshold decoupling and axion-current anomaly-matching map, followed by quality and electric-dipole-moment control. That map remains open.

## 6 Retired Calabi–Yau benchmark

The former values  $\zeta_2/\zeta_1 \simeq 0.560$ ,  $\zeta_3/\zeta_1 \simeq 0.229$ , their derived Kahler ratios, the fitted internal volume, exceptional-cycle coefficients, and corresponding axion decay constants all depended on the withdrawn gauge profile. They are not updated by substituting new ratios into the old formulas because topology, stability, warping, normalization, and degeneracy were not independently selected.

A future geometric execution must publish:

1. the compactification topology, intersection form, bundle, flux and stability chamber;
2. all priors, constraints, discrete branches, flat directions and degeneracies;
3. the absolute-unit bridge and threshold scheme; and
4. a held-out comparison not used to select the geometry.

## 7 Conclusion

Execution I now has a clean split. Gauge/precision profile execution, the selected rank-two literal Cech–HYM witness, and the internal  $K_{\text{threshold}}$  machinery are closed at their declared standards. Physical threshold/mass values remain admitted profile rows, and the old Calabi–Yau numerical

lift is retired. Axion ratios and a conditional PQ lane survive, while absolute normalization and strong-CP selection remain open.

The result supports embedded renormalized-SM equivalence at the adopted profile standard. It does not establish a zero-knob compactification, derive the measured gauge rows, or solve strong CP.

## References

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